

1. Calculate the total resistance in Ohms,

Series

$$\begin{aligned} R &= R_1 + R_2 + R_3 \\ &= 20 + 30 + 50 \\ &= \underline{\underline{100\Omega}} \end{aligned}$$

Parallel

$$\begin{aligned} \frac{1}{R} &= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} & \frac{1}{R} &= \frac{2}{25} \\ \frac{1}{R} &= \frac{1}{20} + \frac{1}{100} + \frac{1}{50} = \frac{2}{25} & R &= \underline{\underline{12.5\Omega}} \end{aligned}$$

2- Calculate the total current from the power supply

Series

$$\begin{aligned} I &= \frac{V}{R} = \frac{125}{100} \\ &= \underline{\underline{1.25}} \end{aligned}$$

Parallel

$$\begin{aligned} I &= \frac{V}{R} = \frac{125}{12.5} \\ &= \underline{\underline{10}} \end{aligned}$$

3. Calculate the current through each resistor

Series

$$I = \frac{V}{R} = \frac{125}{100} = \underline{\underline{1.25}}$$

Parallel

$$I = \frac{V}{R} \quad \frac{125}{20} = \underline{\underline{6.25}} \quad \frac{125}{100} = \underline{\underline{1.25}} \quad \frac{125}{50} = \underline{\underline{2.5}}$$

4. Calculate the Voltage drop through each resistor

Series

$$V = IR$$

$$= 1.25(20)$$

$$= 1.25(30)$$

$$= 1.25(50)$$

$$= \underline{25}$$

$$= \underline{37.5}$$

$$= \underline{62.5}$$

Parallel

= Voltage supplied by power source

$$V = \underline{125}$$

5. Power dissipated through each resistor

Series

$$P = VI$$

$$25(1.25) = 31.25$$

$$37.5(1.25) = 46.875$$

$$62.5(1.25) = 78.125$$

Parallel

$$P = VI$$

$$125(6.25) = 781.25$$

$$125(1.25) = 156.25$$

$$125(2.5) = 312.5$$